



Bangladesh University of Business and Technology (BUBT) Faculty of Engineering & Applied Sciences (FEAS) Department of Computer Science and Engineering (CSE)

LAB MANUAL

1	Program	B.Sc. Eng. in CSE													
2	Course Code	CSE 206													
3	Course Title	Digital Logic Design Lab													
4	Course Type	Core Course													
5	Credit Hour	1.50													
6	Course Objectives	The main objectives of this sessional course are to reinforce the theory presented in the CS Digital Logic Design course by allowing students to experiment with digital logic in a ha manner. And also understand both logical and electrical characteristics of digital design.													
7	Text Book	1. Digital Fundamentals, 9 th Edition, by Thomas L. Floyd													
8	Reference Book	1. Digital Design -With an introduction to the Verilog HDL, 5th Edition, by M. Morris Michael D. Ciletti													
9	Course Outcomes (COs)	Upon completing this course students will be able to: CO1: Demonstrate the working principles of different ICs & basic logic gates. CO2: Apply different IC's & logic gates for different digital logic circuits. CO3: Design and implement combinational circuits and sequential circuits.													
Mapping of COs to POs		Cours e Outco me	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO1 1	PO1 2	
		CO1	√												
		CO2									√				
		CO3			√										
CO No.		PO No.	Bloom's Domain / Level		Delivery Methods / Activities					Assessment Tools					
CO1		PO1	Cognitive / Understanding		Lectures, Lab Task, Assignment.					Lab Performance					
CO2		PO9	Affective/Respondin g		Lectures, Lab Task, Assignment.					Lab Performance and Final Exam					
CO3		PO3	Psychomotor / Guided Response		Lectures, Lab Task, Assignment.					Lab Performance and Final Exam					
10 Weekly Schedule															
Week		Lab	Task										Outcom e		
Week1		Lab1	To verify the Behavior of Logic Gates using Truth Table and familiarization with Digital Integrated Circuits										CO1		

Week2	Lab2	Implementation of Boolean Function using Logic Gates Lab Performance Evaluation	CO1
Week3	Lab3	Simplification of Boolean function by using Boolean Algebra and K- Map. Lab Performance Evaluation	CO2
Week4	Lab4	Construct and test various adder and sub-tractor circuits Lab Performance Evaluation	CO2
Week5	Lab5	Design, construct and test combination logic circuits like Majority Logic circuit, Parity Generator etc.	CO2
Week6	Lab6	Design of Code converters like Gray Code to Binary, Seven-Segment Display etc. Lab Performance Evaluation	CO2
Week7	Lab7	To check the operation of 2-to-4-line Decoder/3-to-8-line Decoder and design 4 to 16 line decoder.	CO2
Week8	Mid Term Week		
Week9	Lab8	To Design BCD (Binary Code Decimal) to 7-SEGMENT Decoder.	CO2
Week10	Lab9	To check the operation of active low decoder. Binary Coded Decimal encoder. Designing of octal to binary encoder and decimal to binary encoder. Lab Performance Evaluation	CO2
Week11	Lab10	Designing of 2-to-1, 4-to-1-line multiplexer and a quadruple 2-to-1-line MUX. Also implement 1-to-4 line de-multiplexer.	CO3
Week12	Lab11	Designing of 4-BIT Magnitude comparator and magnitude comparison using 74LS85 IC Implementing a Full Adder using (a) Decoder (b) Multiplexer	CO3
Week13	Lab12	Construct, test and investigate the operation of various flip-flop circuits, SR Latch, RS flip-flops, D flip-flop etc. Lab Performance Evaluation	CO3
Week 14	Lab13	Construction of Asynchronous Counters, Synchronous Counters and Registers. Lab Performance Evaluation	CO3
Week15	Lab Final Exam		

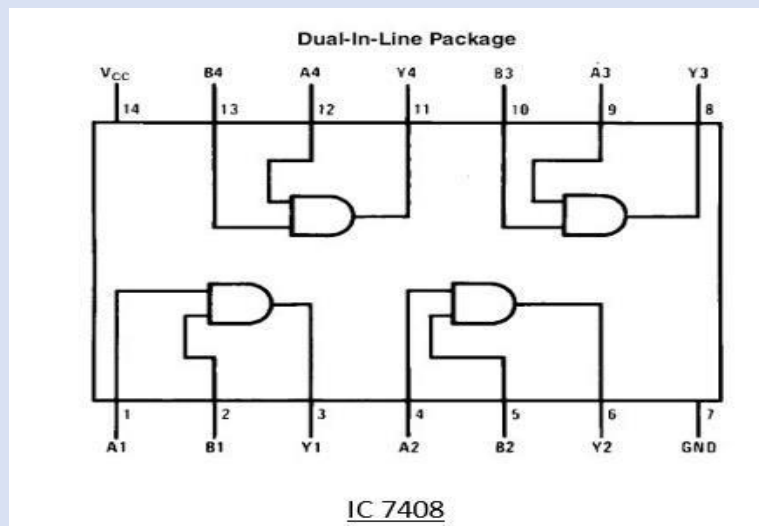
11	S/W and H/W Requirements	The following software's are needed to conduct this lab course: (i) Proteus (ii) Circuitverse The following equipment's are needed to conduct this lab course: (i) Breadboard (ii) IC's (iii) Connecting wires (iv) LED's																																													
12	Experiment No.: 01	<u>Experiment No. 01:</u> <u>Name of the Experiment:</u> To verify the Behavior of Logic Gates using Truth Table and familiari with Digital Integrated Circuits. <u>Objective:</u> Verification and interpretation of truth tables for AND, OR, NOT, NAND, NOR Exclusive OR (EX-OR), Exclusive NOR (EX-NOR) Gates. <u>Course Outcomes (COs), Program Outcomes (POs) and Assessment:</u> <table><tr><th>CO Statement</th><th>Correspondin g PO</th><th>Domain / level of learning taxonomy</th><th>Delivery methods and activities</th><th>Assessme tools</th></tr><tr><td>CO1: Demonstrate the working principles of different ICs & basic logic gates.</td><td>Engineering Knowledge (PO1)</td><td>Cognitive / Understanding</td><td>☒ Simulation ☒ n Experiment ☒ Practice lab ☒ Group discussion</td><td>☒ Lab tes ☒ Lab reports ☒ Final l test ☐ Open ended lab ☐ Project show & project presentation</td></tr></table> <u>Theory:</u> Logic gates are electronic circuits which perform logical functions on one or more inputs to pr one output. There are seven logic gates. When all the input combinations of a logic gate are wri a series and their corresponding outputs written along them, then this input output combination is Truth Table. OR, AND, NOT are basic gates. NAND, NOR are known as universal gates. Variou and their working is explained here. <u>Required equipment and devices:</u> <table><tr><th>SL No.</th><th>COMPONENT</th><th>SPECIFICATION</th><th>QTY</th></tr><tr><td>1.</td><td>AND GATE</td><td>IC 7408</td><td>1</td></tr><tr><td>2.</td><td>OR GATE</td><td>IC 7432</td><td>1</td></tr><tr><td>3.</td><td>NOT GATE</td><td>IC 7404</td><td>1</td></tr><tr><td>4.</td><td>NAND GATE 2 I/P</td><td>IC 7400</td><td>1</td></tr><tr><td>5.</td><td>NOR GATE</td><td>IC 7402</td><td>1</td></tr><tr><td>6.</td><td>X-OR GATE</td><td>IC 7486</td><td>1</td></tr><tr><td>7.</td><td>IC TRAINER KIT</td><td></td><td>1</td></tr></table>				CO Statement	Correspondin g PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessme tools	CO1: Demonstrate the working principles of different ICs & basic logic gates.	Engineering Knowledge (PO1)	Cognitive / Understanding	☒ Simulation ☒ n Experiment ☒ Practice lab ☒ Group discussion	☒ Lab tes ☒ Lab reports ☒ Final l test ☐ Open ended lab ☐ Project show & project presentation	SL No.	COMPONENT	SPECIFICATION	QTY	1.	AND GATE	IC 7408	1	2.	OR GATE	IC 7432	1	3.	NOT GATE	IC 7404	1	4.	NAND GATE 2 I/P	IC 7400	1	5.	NOR GATE	IC 7402	1	6.	X-OR GATE	IC 7486	1	7.	IC TRAINER KIT		1
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BASIC GATES:

1. AND GATE

The AND gate performs a logical multiplication commonly known as AND function. The output is high only when both the inputs are high. The output is low level when any one of the inputs is low.

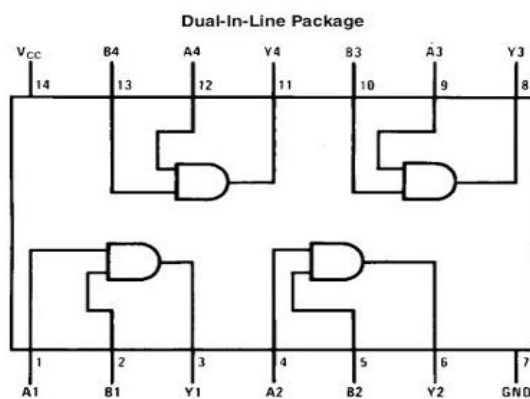
InputA	InputB	Output
1	1	1
1	0	0
0	1	0
0	0	0



2. OR GATE

The OR gate performs a logical addition commonly known as OR function. The output is high if any one of the inputs is high. The output is low level when both the inputs are low.

InputA	InputB	Output
1	1	1
1	0	1
0	1	1
0	0	0

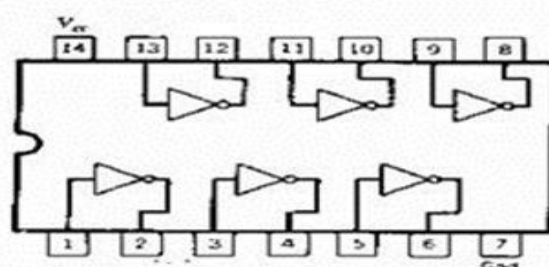


IC 7432

3. NOT GATE

The NOT gate is called an inverter. The output is high when the input is low. The output is low when the input is high.

Input	Output
0	1
1	0



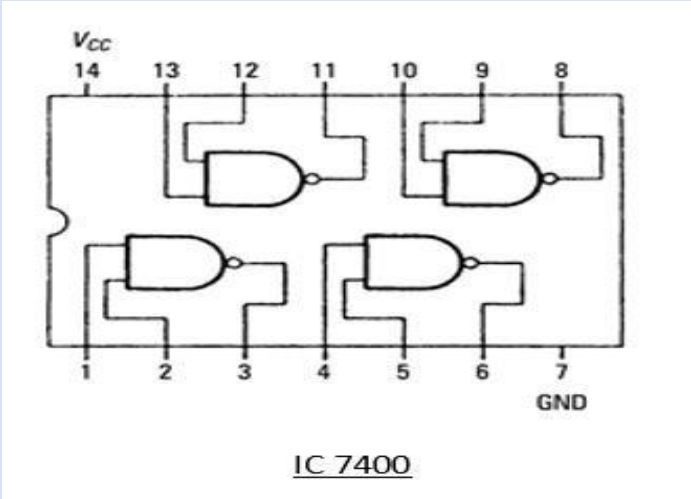
IC 7404

UNIVERSAL GATES:

1. NAND GATE

The NAND gate is a contraction of AND-NOT. The output is high when both inputs are low and one of the inputs is low. The output is low level when both inputs are high.

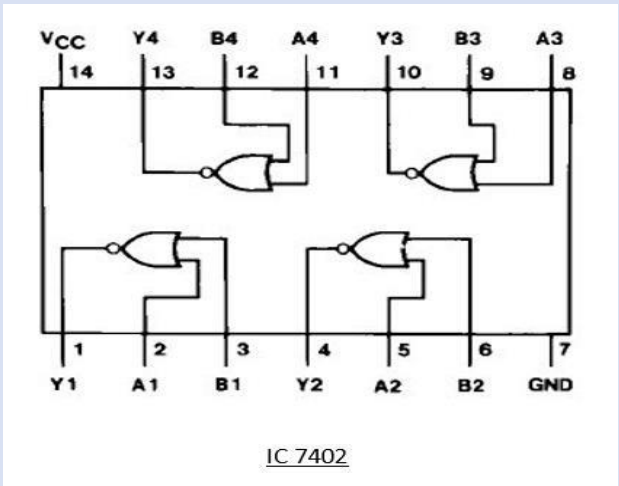
Input A	Input B	Output
1	1	0
1	0	1
0	1	1
0	0	1



2. NOR GATE

The NOR gate is a contraction of OR-NOT. The output is high when both inputs are low. The output is low when one or both inputs are high.

Input A	Input B	Output
1	1	0
1	0	0
0	1	0
0	0	1



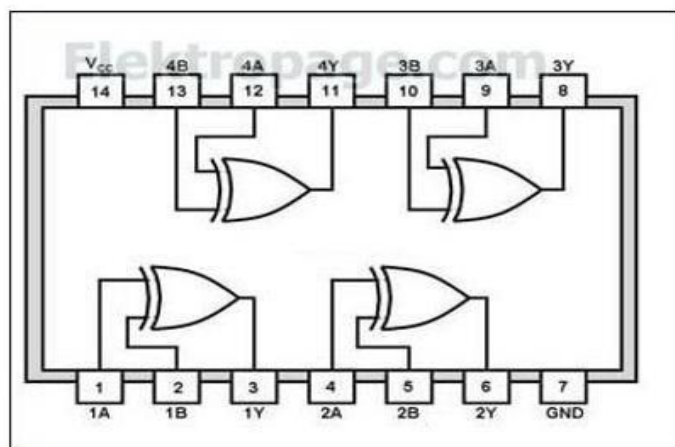
ADVANCED GATES:

1. X-OR GATE

The output is high when any one of the inputs is high. The output is low when both the inputs are low and both the inputs are high.

Input A	Input B	Output
1	1	0

1	0	1
0	1	1
0	0	0



IC 74136

PROCEDURE:

1. Connect the trainer kit to ac power supply.
2. Connect the inputs of any one logic gate to the logic sources and its output to the log indicator.
3. Apply various input combinations and observe output for each one.
4. Verify the truth table for each input/ output combination.
5. Repeat the process for all other logic gates.
6. Switch off the ac power supply.

DISCUSSIONS:

References:

1. Reference book.
2. Online sources.

2 Experiment
1 No.: 02

Experiment No.02

Name of the Experiment: Realization of Basic gates using Universal Gates.

- **Objective:** To construct NOT, AND, OR, Exclusive OR (EX-OR), Exclusive NOR (E NOR) logic gates using only NAND gates.
- To construct NOT, AND, OR, Exclusive OR (EX-OR), Exclusive NOR (EX-NOR) logic gates using only NOR gates.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1: Demonstrate the working principles of different ICs & basic logic gates.	Engineering Knowledge (PO1)	Cognitive / Understanding	☒ Simulation ☒ Experiment ☒ Practice ☒ Lab ☒ Group discussion ☒ Tutorial	☒ Lab tests Lab reports Final ☒ lab ☒ test ☐ Open ended lab ☐ Project show & project presentation ☐

Components required:

SL No.	COMPONENT	SPECIFICATION	QTY
1.	NAND GATE 2 I/P	IC 7400	1
2.	NOR GATE 2 I/P	IC 7402	1
3.	IC TRAINER KIT	-	1
4.	PATCH CORD	-	As per required

(A) NAND AS A UNIVERSAL GATE:

THEORY:

NAND gate is actually a combination of two logic gates: AND gate followed by NOT gate. So its output is complement of the output of an AND gate. This gate can have minimum two inputs, output is a one. By using only NAND gates, we can realize all logic functions: AND, OR, NOT, X-OR, X-NOR. So this gate is also called universal gate.

1. NAND gate as NOT gate:

A NOT produces complement of the input. It can have only one input, tie the inputs of a NAND gate together. Now it will work as a NOT gate. Its output is

$$Y = (A.A)' = (A)'$$

2. NAND gates as AND gate:

A NAND produces complement of AND gate. So, if the output of a NAND gate is inverted, overall output will be that of an AND gate.

$$Y = ((A.B)')' = (A.B)$$

3. NAND gates as OR gate:

From De Morgan's theorems: $(A.B)' = A' + B'$. Similarly, $(A'.B')' = A'' + B'' = A + B$. So, give the inverted inputs to a NAND gate, obtain OR operation at output.

4. NAND gates as EX-OR gate:

The output of a two input EX-OR gate is given by: $Y = A'B + AB'$. EX-OR gate can be implemented using four NAND gates as follows.

Gate No.	Inputs	Output
1	A, B	$(AB)'$
2	A, $(AB)'$	$(A(AB)')'$
3	$(AB)'$, B	$(B(AB)')'$
4	$(A(AB)')'$, $(B(AB)')'$	$A'B + AB'$

Now the output from gate no. 4 is the overall output of the configuration.

$$\begin{aligned}
 Y &= ((A(AB)')' (B(AB)')')' \\
 &= (A(AB)')'' + (B(AB)')'' \\
 &= (A(AB)') + (B(AB)') \\
 &= (A(A' + B')) + (B(A' + B')) \\
 &= (AA' + AB') + (BA' + BB') \\
 &= (0 + AB' + BA' + 0) \\
 &= AB' + BA'
 \end{aligned}$$

So $Y = AB' + A'B$

5. NAND gates as EX-NOR gate

EX-NOR gate is actually EX-OR gate followed by NOT gate. So give the output of EX-OR gate to a NOT gate, overall output is that of an EX-NOR gate.

$$Y = AB + A'B'$$

PROCEDURE:

- (i) Verify the gates and connect the NAND gates as per logic diagrams (A) for any of the logic functions to be realized.
- (ii) Connect Pin-14 of all ICs to +5V and Pin-7 to ground.
- (iii) Feed the logic 0 (0V) or 1(5V) in different combinations at the inputs A & B according to truth table.
- (iv) Observe and note down the output readings for Y for different combinations of inputs and verify the truth table for input/output combination
- (v) Repeat the process for all logic functions/gates.

Logic diagrams (A) observations table (A)

1. NAND gate as NOT gate:

A	Y	
0	1	
1	0	

2. NAND gates as AND gate:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

3. NAND gates as OR gate:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

4. NAND gates as EX-OR gate:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

5. NAND gates as EX-NOR gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

DISCUSSIONS (A):

(B) NOR AS A UNIVERSAL GATE:

THEORY:

NOR gate is actually a combination of two logic gates: OR gate followed by NOT gate. So its output is complement of the output of an OR gate. This gate can have minimum two inputs, output is always one. By using only NOR gates, we can realize all logic functions: AND, OR, NOT, X-OR, X-NAND. So this gate is also called universal gate.

1. NOR gate as NOT gate:

A NOT produces complement of the input. It can have only one input, tie the inputs of a NOR together. Now it will work as a NOT gate. Its output is

$$Y = (A+A)' = (A)'$$

2. NOR gates as OR gate:

A NOR produces complement of OR gate. So, if the output of a NOR gate is inverted, overall output will be that of an OR gate.

$$Y = ((A+B)')' = (A+B)$$

3. NOR gates as AND gate:

From De Morgan's theorems: $(A+B)' = A' \cdot B'$. Similarly, $(A'+B')' = A'' \cdot B'' = A \cdot B$. So, give the inverted inputs to a NOR gate, obtain AND operation at output.

4. NOR gates as EX-NOR gate:

The output of a two input EX-NOR gate is given by: $Y = AB + A'B'$. EX-NOR gate can be implemented using four NOR gates as follows.

Gate No.	Inputs	Output
1	A, B	$(A + B)'$
2	A, $(A + B)'$	$(A + (A+B)')'$
3	$(A + B)'$, B	$(B + (A+B)')'$
4	$(A + (A + B)')'$, $(B + (A+B)')'$	$AB + A'B'$

Now the output from gate no. 4 is the overall output of the configuration.

$$\begin{aligned}
 Y &= ((A + (A+B)')' (B + (A+B)')')' \\
 &= (A + (A+B)')'' \cdot (B + (A+B)')'' \\
 &= (A + (A+B)') \cdot (B + (A+B)') \\
 &= (A + A'B') \cdot (B + A'B') \\
 &= (A + A') \cdot (A + B') \cdot (B + A') \cdot (B + B') \\
 &= 1 \cdot (A+B') \cdot (B+A') \cdot 1 \\
 &= (A+B') \cdot (B+A') \\
 &= A \cdot (B + A') + B' \cdot (B+A') \\
 &= AB + AA' + B'B + B'A' \\
 &= AB + 0 + 0 + B'A' \\
 &= AB + B'A'
 \end{aligned}$$

$$\text{So } Y = AB + A'B'$$

5. NOR gates as EX-OR gate

EX-OR gate is actually EX-NOR gate followed by NOT gate. So give the output of EX-NOR to a NOT gate, overall output is that of an EX-OR gate.

$$Y = A'B + AB'$$

PROCEDURE:

- Verify the gates and connect the NOR gates as per logic diagrams (B) for any of the log functions to be realized.
- Connect Pin-14 of all ICs to +5V and Pin-7 to ground.
- Feed the logic 0 (0V) or 1(5V) in different combinations at the inputs A & B according to truth table.

- (iv) Observe and note down the output readings for Y for different combinations of inputs and verify the truth table for input/output combination
- (v) Repeat the process for all logic functions/gates

LOGIC DIAGRAMS (B):
TABLE (

OBSERVATION

1. NOR gate as NOT gate:

A	Y
0	1
1	0

2. NOR gates as OR gate:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

3. NOR gates as AND gate:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

4. NOR gates as EX-NOR gate:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

5. NOR gates as EX-OR gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

22	Experiment No.: 03	DISCUSSIONS (B):																			
		References: 1. Reference book. 2. Online sources.																			
		<u>Experiment No. 03</u>																			
		<u>Name of the experiment:</u> Simplification of Boolean function.																			
		Objectives: To simplify the Boolean expression and to build the logic circuit. Given a Truth table, derive the Boolean expressions and build the logic circuit to realize it.																			
		<u>Course Outcomes (COs), Program Outcomes (POs) and Assessment:</u>																			
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CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools																	
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CO1: Demonstrate the working principles of different ICs & basic logic gates.	Engineering Knowledge (PO1)	Cognitive / Understanding	Simulation☒ Experiment☒ Practice lab☒ Group discussion☒ Tutorial☒	Lab reports☒ Final lab test☒ Open ended lab☒ Project show & project presentation☒																	
		Components Required: IC 7400, IC 7408, IC 7432, IC 7406, IC 7402, IC 7404, IC 7486																			
		Theory: Canonical Forms (Normal Forms): Any Boolean function can be written in disjunctive normal form (sum of min-terms) or conjunctive normal form (product of max-terms). A Boolean function can be represented by a Karnaugh map in which each cell corresponds to a min-term. The cells are arranged in such a way that any two immediately adjacent cells correspond to two min-terms at distance 1. There is more than one way to construct a map with this property.																			

Truth Table:

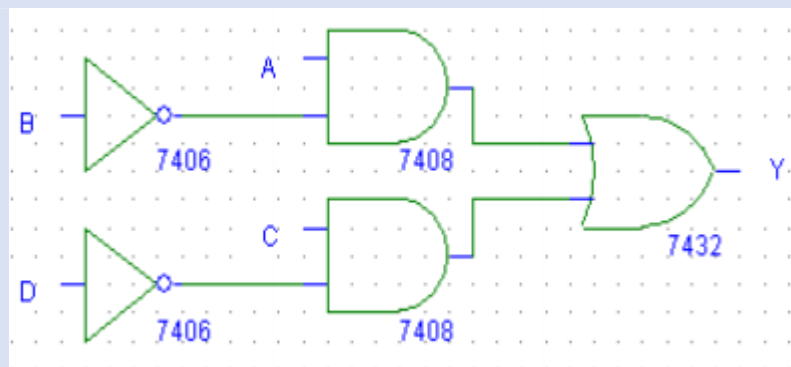
1) $Y = \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}CD$

AB

			1
			1
			1
1	1	1	1

After simplifying using K-Map method we get $Y = A\bar{B} + C\bar{D}$

INPUTS				OUTPUT
A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

Logic Diagram:**Procedure:**

1. Check the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.

4. Provide the input data via the input switches and observe the output on output LEDs

Discussion:

References:

1. Reference book.
2. Online sources.

23 Experiment
No.: 04

Experiment No. 04

Name of the experiment: Construct and test various adder and subtractor circuits.

- **Objective:** To realize the adder and subtractor circuits using basic gates and universal
- To realize full adder using two half adders.
- To realize a full subtractor using two half subtractors.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2: Apply different IC's and logic gates for different digital logic circuits	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Components Required:

IC 7400, IC 7408, IC 7486, IC 7432, Patch Cords & IC Trainer Kit.

Theory:

Half-Adder: A combinational logic circuit that performs the addition of two data bits, A and B, called a half-adder. Addition will result in two output bits; one of which is the sum bit, S, and the other is the carry bit, C. The Boolean functions describing the half-adder are:

$$S = A \oplus B \quad C = A B$$

Full-Adder: The half-adder does not take the carry bit from its previous stage into account.

This carry bit from its previous stage is called carry-in bit. A combinational logic circuit that adds two data bits, A and B, and a carry-in bit, C_{in} , is called a full-adder. The Boolean functions describing the full-adder are:

$$S = (x \oplus y) \oplus C_{in} \quad C = xy + C_{in}(x \oplus y)$$

Half Subtractor: Subtracting a single-bit binary value B from another A (i.e. $A - B$) produces a difference bit D and a borrow out bit B-out. This operation is called half subtraction and the circuit to realize it is called a half subtractor. The Boolean functions describing the half Subtractor are:

$$S = A \oplus B \quad C = A'B$$

Full Subtractor: Subtracting two single-bit binary values, B, C_{in} from a single-bit value A produces a difference bit D and a borrow out Br bit. This is called full subtraction. The Boolean functions describing the full-subtractor are:

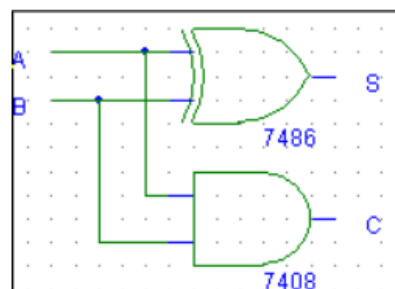
$$D = (x \oplus y) \oplus C_{in} \quad Br = A'B + A'(C_{in}) + B(C_{in})$$

Truth Table & Logic Diagram : To realize Half Adder

TRUTH TABLE

INPUTS		OUTPUTS	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

i) Basic Gates

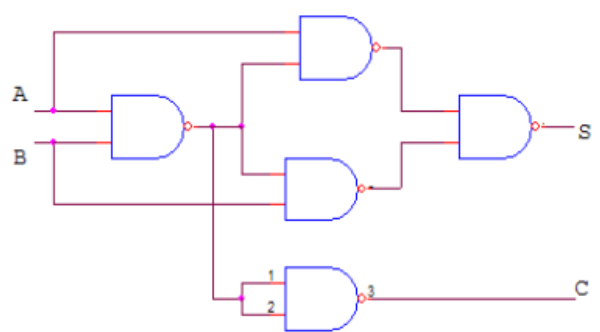


BOOLEAN EXPRESSIONS:

$$S = A \oplus B$$

$$C = A B$$

ii) NAND Gates



Full Adder

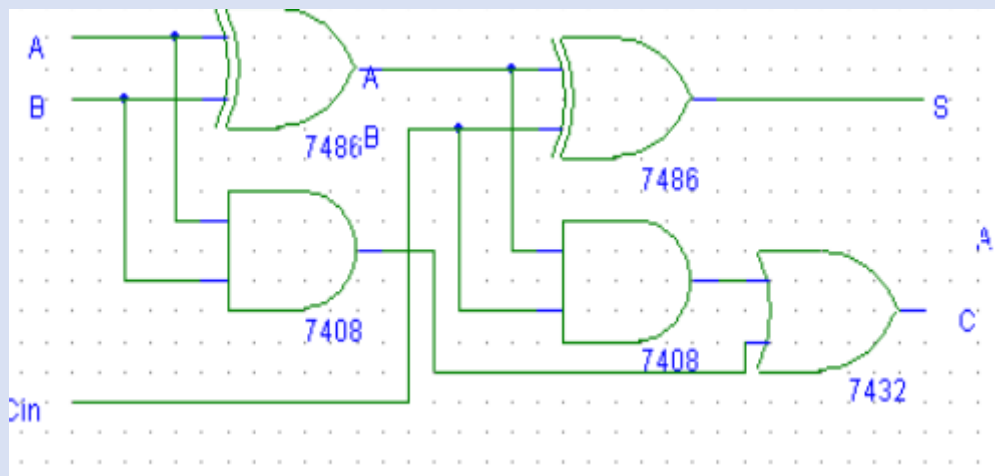
TRUTH TABLE

INPUTS			OUTPUTS	
A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

BOOLEAN EXPRESSIONS:

$$S = A \oplus B \oplus C$$

$$C = A B + B \text{ Cin} + A \text{ Cin}$$

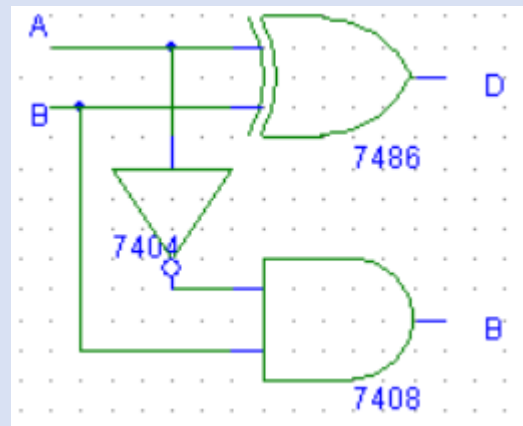
**HALF SUBTRACTOR****TRUTH TABLE**

INPUTS		OUTPUTS	
A	B	D	Br
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

BOOLEAN EXPRESSIONS

$$D = A \oplus B$$

$$\text{Br} = \bar{A} B$$



FULL SUBTRACTOR

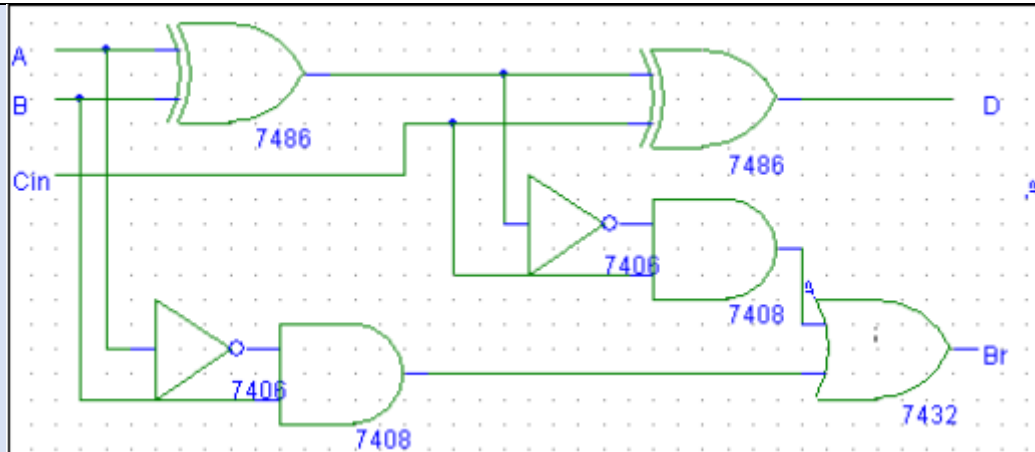
TRUTH TABLE

INPUTS			OUTPUTS	
A	B	Cin	D	Br
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

BOOLEAN EXPRESSIONS:

$$D = A \oplus B \oplus C$$

$$Br = \bar{A} B + B Cin + \bar{A} Cin$$



1. Check the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.
4. Provide the input data via the input switches and observe the output on output LEDs

References:

1. Reference book.
2. Online sources.

Experiment No. 05

Name of the experiment: Parity generator and Checker.

Objective: To design and verify the truth table of a three-bit odd parity generator and checker

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2: Apply different IC's and logic gates for different digital logic circuits	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment ☒ Practice lab	☒ Lab tests ☒ Lab reports ☒ Final lab test

☒ Group discussion
☒ Tutorial

☐ Open ended lab
☐ Project show & project presentation
COMPONENTS REQUIRED:

IC Trainer kit, ICS 7400, IC 7486

THEORY:

A parity bit is used for the purpose of detecting errors during transmission of binary information. A parity bit is an extra bit included with a binary message to make the number of 1's either odd or even. The message including the parity bit is transmitted and then checked at the receiving end for errors. An error is detected if the checked parity does not correspond with the one transmitted. The circuit that generates the parity bit in the transmitter is called a parity generator and the circuit that checks the parity in the receiver is called a parity checker. In even parity the added parity bit will make the total number of 1's an even amount and in odd parity the added parity bit will make the total number of 1's an odd amount. In a three-bit odd parity generator the three bits in the message together with the parity bit are transmitted to their destination, where they are applied to the parity checker circuit. The parity checker circuit checks for possible errors in the transmission. Since the information was transmitted with odd parity the four bits received must have an odd number of 1's. An error occurs during the transmission if the four bits received have an even number of 1's, indicating that one bit has changed. The output of the parity checker is denoted by PEC (parity error check) and it is equal to 1 if an error occurs, i.e., if the four bits received have an even number of 1's.

TRUTH TABLE:

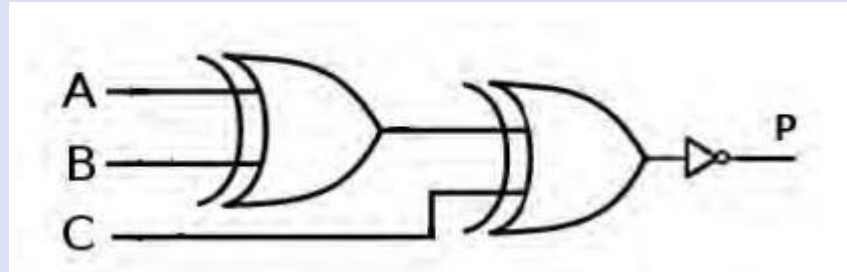
INPUTs (Three bit messages)			OUTPUTs (Odd Parity bit)
A	B	C	P
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

From the truth table the expression for the output parity bit is, $P(A, B, C) = \sum m(0, 3, 5, 6)$ Also written as,

$$P = A'B'C' + A'BC + AB'C + ABC'$$

$$P = (A \oplus B \oplus C)'$$

CIRCUIT DIAGRAM: Odd Parity Generator



ODD PARITY CHECKER

INPUT (four bit messageReceived)				OUTPUT (Parity error check)
A	B	C	P	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

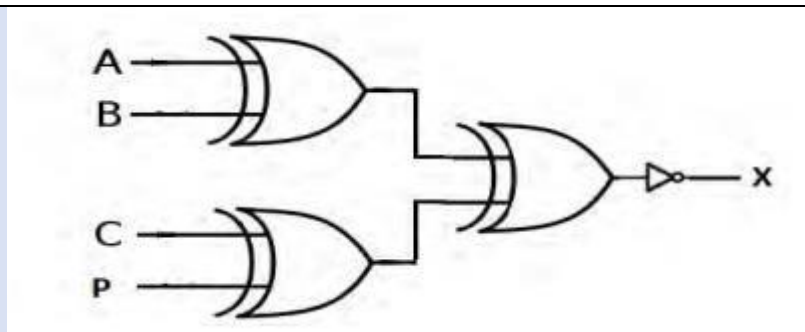
From the truth table the expression for the output parity checker bit is,

$$X(A, B, C, P) = \sum (0, 3, 5, 6, 9, 10, 12, 15)$$

The above expression is reduced as,

$$X = (A \oplus B \oplus C \oplus P)'$$

CIRCUIT DIAGRAM: Odd Parity Checker

**PROCEDURE:**

- Connections are given as per the circuit diagrams.
- For all the ICs 7th pin is grounded and 14th pin is given +5 V supply.
- Apply the inputs and verify the truth table for the Parity generator and checker.

DISCUSSION:

The design of the three-bit odd Parity generator and checker circuits was done and their truth were verified.

References:

1. Reference book.
2. Online sources.

25 **Experiment No.: 06**

Experiment No. 06

Name of the Experiment: BCD TO EXCESS- 3 CODE CONVERTERS.

Objective: To learn to realize BCD to Excess-3 code using adder IC 7483. To learn to realize Excess-3 Code using adder IC 7483.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2: Apply different IC's and logic gates for different digital logic circuits	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show &

COMPONENTS REQUIRED:

IC 7483, IC 7486, Patch Cords & IC Trainer Kit.

THEORY:

Code converter is a combinational circuit that translates the input code word into a new corresponding word. The excess-3 code digit is obtained by adding three to the corresponding BCD digit. To Construct a BCD-to-excess-3-code converter with a 4-bit adder feed BCD code to the 4-bit adder as the first operand and then feed constant 3 as the second operand. The output is the corresponding excess-3 code. To make it work as an excess-3 to BCD converter, we feed excess-3 code as the first operand and then feed 2's complement of 3 as the second operand. The output is the BCD code.

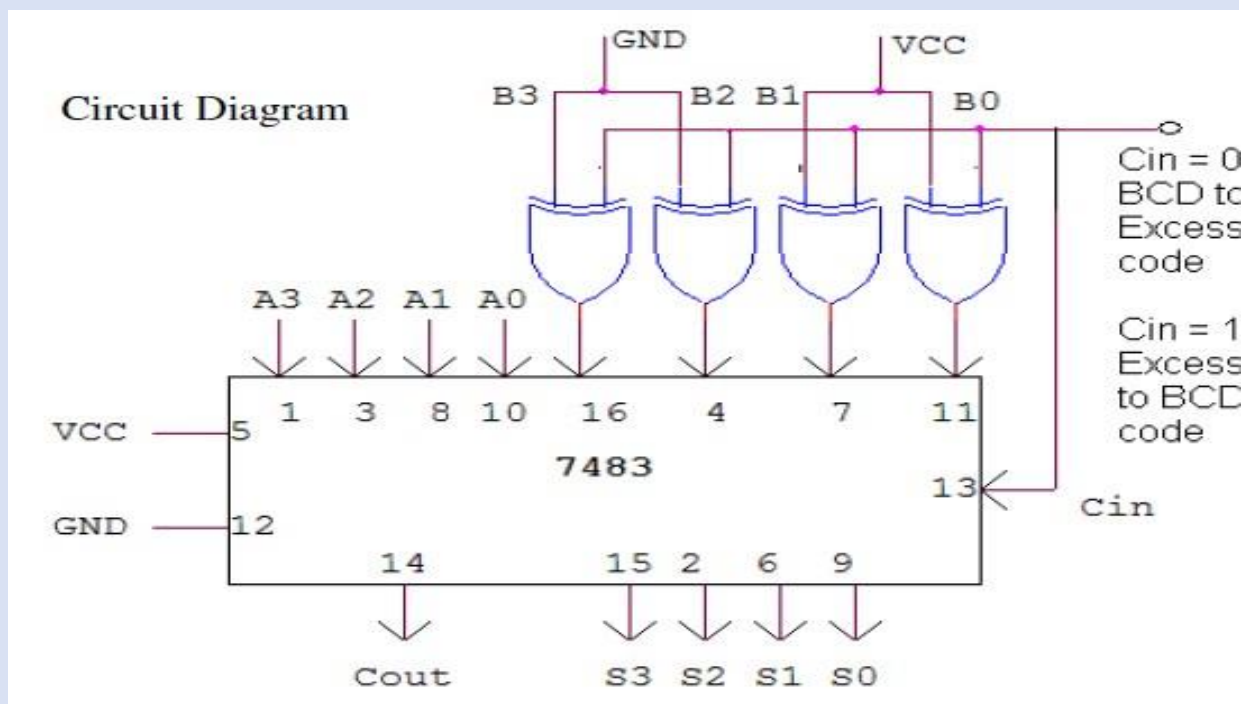
TRUTH TABLE:**i) BCD - EXCESS-3 CODE**

BCD				Excess-3			
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0

ii) EXCESS-3 – BCD CODE

Excess-3				BCD			
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0

1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1



PROCEDURE:

- Check all the components for their working.
- Insert the appropriate IC into the IC base.
- Make connections as shown in the circuit diagram.
- Apply BCD code as first operand(A) and binary 3 as second operand(B) and cin=0 for realizing BCD-to-Excess-3-code:
- Apply Excess-3-code code as first operand (A) and binary 3 as second operand (B) and cin=1 for realizing Excess-3-code to BCD.
- Verify the Truth Table and observe the outputs.

DISCUSSION:

Realized BCD code to Excess-3 code conversion and vice versa using 7483 IC

References:

1. Reference book.
2. Online sources.

26 Experiment No.: 07

Experiment No. 07

Name of the Experiment: To check the operation of 2-to-4 line Decoder.

Objective: To learn about working principle of decoder. To learn and understand the working of IC 74139. To realize using basic gates as well as universal gates.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2: Apply different IC's and logic gates for different digital logic circuits	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Components Required:

- IC74LS139, IC 7400, IC 7408, IC 7432, IC 7404, IC 7410.
- Patch chords.
- IC Trainer Kit.

Theory:

A decoder is a combinational circuit that connects the binary information from 'n' input line maximum of 2^n unique output lines. Decoder is also called a min-term generator/max-term generator. A min-term generator is constructed using AND and NOT gates. The appropriate output is indicated by logic 1 (positive logic). Max-term generator is constructed using NAND gates. The appropriate output is indicated by logic 0 (Negative logic).

The IC 74139 accepts two binary inputs and when enable provides 4 individual active low output device has 2 enable inputs (Two active low).

▪ **2:4 DECODER (MIN TERM GENERATOR):**

Truth Table:

INPUT		OUTPUT			
A	B	Y0	Y1	Y2	Y3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Boolean Expression:

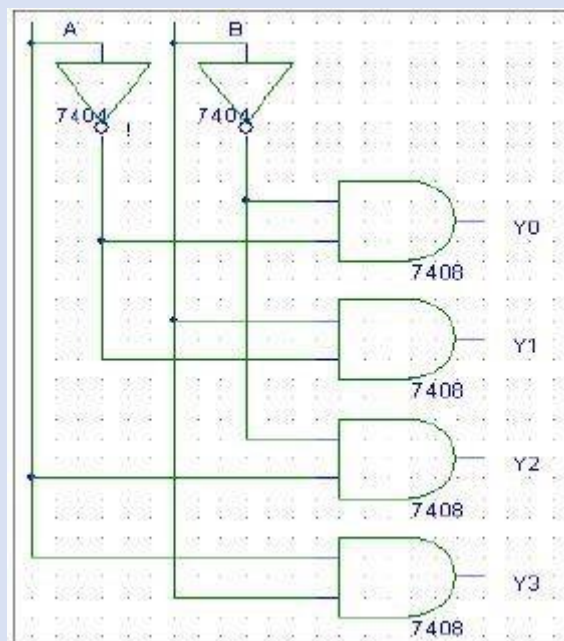
$$Y0 = \overline{A}\overline{B}$$

$$Y1 = \overline{A}B$$

$$Y2 = A\overline{B}$$

$$Y3 = AB$$

Circuit Diagram:

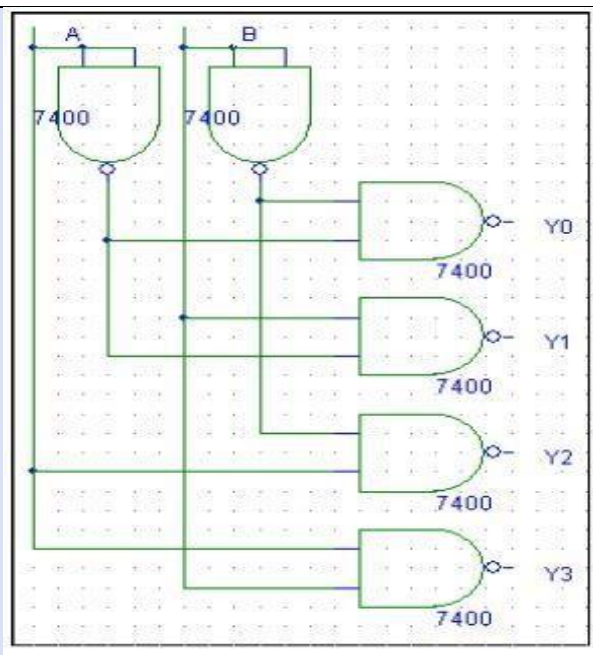


▪ **2:4 DECODER (MAX TERM GENERATOR):**

Truth Table:

INPUT		OUTPUT			
A	B	Y0	Y1	Y2	Y3
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0

Circuit Diagram:

**Procedure:**

- Make the connections as per the circuit diagram.
- Change the values of G1, G2A, G2B, A, B, and C, using switches.
- Observe status of Y0, to Y7 on LED's.
- Verify the truth table.

Discussions:

Verified the Operation of 2 to 4 Decoder.

References:

1. Reference book.
2. Online sources.
1. .

27 Experiment
No.: 08

Experiment No. 08

Name of the Experiment: Using the 7447 IC, design a BCD (Binary Code Decimal) to 7-SEGMENT Decoder.

Objective: To learn about various applications of decoder. To learn about types of seven-segment d

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2:Apply different IC's and logic gates for different	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment	☒ Lab tests ☒ Lab reports

digital logic
circuits
☒ Practice lab
☒ Group discussion
☒ Tutorial

☒ Final lab test
☐ Open ended lab
☒ Project show & project presentation

Components Required:

- IC7447.
- 7-Segment display (common anode).
- Patch chords, resistor (1K).
- IC Trainer Kit.

Theory:

The Light Emitting Diode (LED) finds its place in many applications in these modern electronic devices. One of them is the Seven Segment Display. Seven-segment displays contain the arrangement of LEDs in “Eight” (8) positions, and a Dot (.) with a common electrode, lead (Anode or Cathode). The purpose of arranging it in that position is that we can make any number out of that by switching ON or OFF the particular LED's. Here is the block diagram of the Seven Segment LED arrangement. LED's are basically of two types:

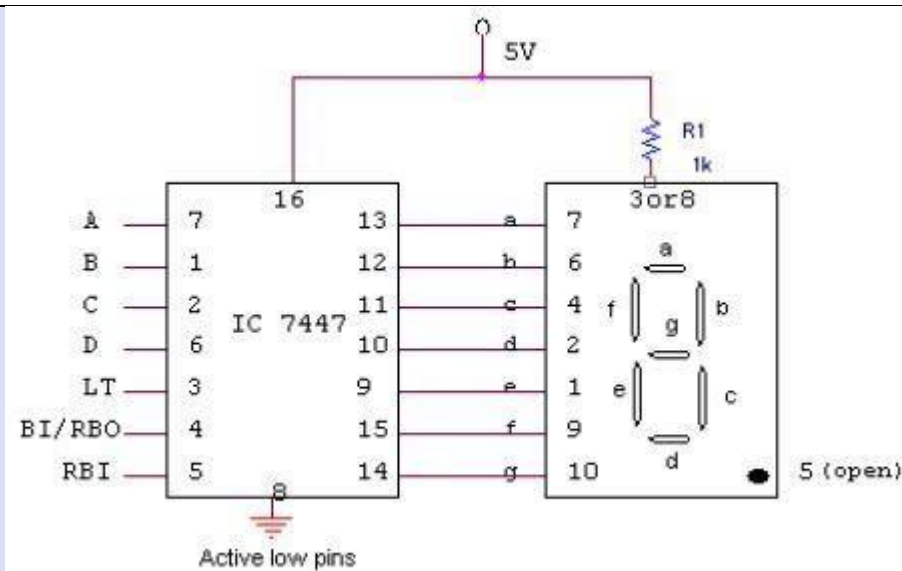
- Common Cathode (CC) -All the 8 anode legs use only one cathode, which is common.
- Common Anode (CA)-The common leg for all the cathode is of Anode type.

A decoder is a combinational circuit that converts the binary information from ‘n’ input lines into a maximum of 2^n unique output lines. The IC7447 is a BCD to 7-segment pattern converter. The IC7447 takes the Binary Coded Decimal (BCD) as the input and outputs the relevant 7 segment code.

Truth Table:

BCD Inputs				Output Logic Levels from IC 7447 to 7-segments							Decimal number display
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	1	0	0	1	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0	2
0	0	1	1	0	0	0	0	1	1	0	3
0	1	0	0	1	0	0	1	1	0	0	4
0	1	0	1	0	1	0	0	1	0	0	5
0	1	1	0	1	1	0	0	0	0	0	6
0	1	1	1	0	0	0	1	1	1	1	7
1	0	0	0	0	0	0	0	0	0	0	8
1	0	0	1	0	0	0	1	1	0	0	9

Circuit Diagram:



Procedure:

- Check all the components for their working.
- Insert the appropriate IC into the IC base.
- Make connections as shown in the circuit diagram.
- Verify the Truth Table and observe the outputs.

Discussions:

It is possible to display the decimal value of a binary number on a 7-segment display using a decoder. However, this method will allow displaying only digits from 0 to 9 and letters A to only way to display number more than 9 is to use a display that has more than 7 segments or just multiple 7-segment displays at once with the corresponding BCD decoder.

References:

3. Reference book.
4. Online sources.

28 Experiment
No.: 09

Experiment No. 09

Name of the experiment: Designing of Decimal to BCD Encoder and Octal to Binary Encoder

Objective: To learn about various applications of Encoders. To learn and understand the work IC 74147 and IC 74148. To learn to do code conversion using encoders.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO2: Apply different IC's and logic gates for	Individual Work and Teamwork (PO9)	Affective/Responding	☒ Simulation ☒ Experiment	☒ Lab tests ☒ Lab reports

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digital logic
circuits
☒ Practice lab
☒ Group discussion
☒ Tutorial

☒ Final lab test
☐ Open ended lab
☐ Project show & project presentation
Components Required:

- IC 74147.
- IC 74148.
- Patch chords.
- IC Trainer Kit.

Theory:

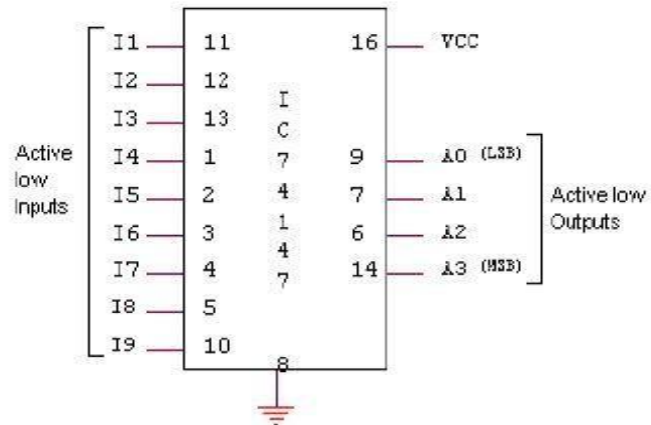
An encoder performs a function that is the opposite of decoder. It receives one or more signals encoded format and output a code that can be processed by another logic circuit. One of the advantages of encoding data, or more often data addresses in computers, is that it reduces the number of required bits to represent data or addresses. For example, if a memory has 16 different locations, in order to access these 16 different locations, 16 lines (bits) are required if the addressing signals are in 1-bit format. However, if we code the 16 different addresses into a binary format, then only 4 lines (bits) are required. Such a reduction improves the speed of information processing in digital systems.

- **Decimal to BCD Encoder**

Truth Table:

INPUTS									OUTPUTS			
I ₁	I ₂	I ₃	I ₄	I ₅	I ₆	I ₇	I ₈	I ₉	A ₃	A ₂	A ₁	A ₀
1	1	1	1	1	1	1	1	0	0	1	1	0
X	X	X	X	X	X	X	0	1	0	1	1	1
X	X	X	X	X	X	0	1	1	1	0	0	0
X	X	X	X	X	0	1	1	1	1	0	0	1
X	X	X	X	0	1	1	1	1	1	0	1	0
X	X	X	0	1	1	1	1	1	1	0	1	1
X	X	0	1	1	1	1	1	1	1	1	0	0
X	0	1	1	1	1	1	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1

Circuit Diagram:

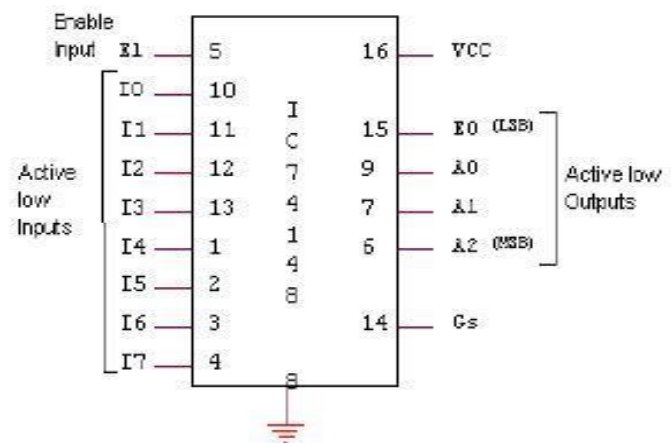


Octal to Binary Encoder

Truth Table:

Inputs									Outputs				
E ₁	I ₀	I ₁	I ₂	I ₃	I ₄	I ₅	I ₆	I ₇	A ₂	A ₁	A ₀	G _S	E ₀
1	X	X	X	X	X	X	X	X	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	0
0	X	X	X	X	X	X	X	0	0	0	0	0	1
0	X	X	X	X	X	X	0	1	0	0	1	0	1
0	X	X	X	X	X	0	1	1	0	1	0	0	1
0	X	X	X	X	0	1	1	1	0	1	1	0	1
0	X	X	X	0	1	1	1	1	1	0	0	0	1
0	X	X	0	1	1	1	1	1	1	0	1	0	1
0	X	0	1	1	1	1	1	1	1	1	0	0	1
0	0	1	1	1	1	1	1	1	1	1	1	0	1

Circuit Diagram:



Procedure:

- Check all the components for their working.
- Insert the appropriate IC into the IC base.

- Make connections as shown in the circuit diagram.
- Verify the Truth Table and observe the outputs.

Discussions:

References:

1. Reference book.
2. Online sources.

29 Experiment No.: 10

Experiment No. 10

Name of the Experiment: To design and set up the following circuit

1. Designing of 2 to 1 , 4 to 1 line multiplexer
2. Designing a quadruple 2 to 1 line MUX
3. Designing 1 to 4 line de-multiplexer.
- 4.

Objective: To learn about various applications of multiplexer and de-multiplexer. To learn understand the working of IC 74153 and IC 74139. To learn to realize any function using Multip

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO3: Design and implement combinational circuit.	Design of Solutions (PO3)	Psychomotor/Guided response	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Components Requirements:

- AND gate (7408), OR gate (7432), NOT gate (7404), IC trainer kit, Wires.

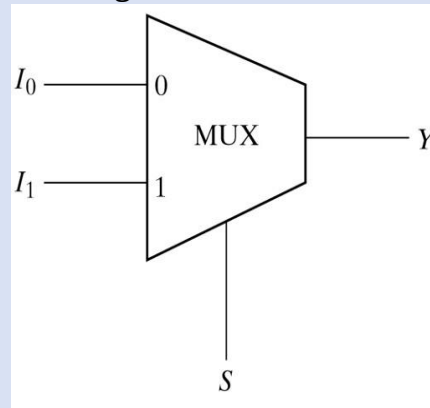
Theory:

Multiplexers are very useful components in digital systems. They transfer a large number of information units over a smaller number of channels, (usually one channel) under the control of selection signals. Multiplexer means many to one. A multiplexer is a circuit with many inputs but only one output.

using control signals (select lines) we can select any input to the output. Multiplexer is also called a data selector because the output bit depends on the input data bit that is selected. The general multiplexer circuit has 2^n input signals, n control/select signals and 1 output signal.

De-multiplexers perform the opposite function of multiplexers. They transfer a small number of information units (usually one unit) over a larger number of channels under the control of select signals. The general de-multiplexer circuit has 1 input signal, n control/select signals and 2^n output signals. De-multiplexer circuit can also be realized using a decoder circuit with enable.

Block Diagram & Truth table:



S	Y
0	I_0
1	I_1

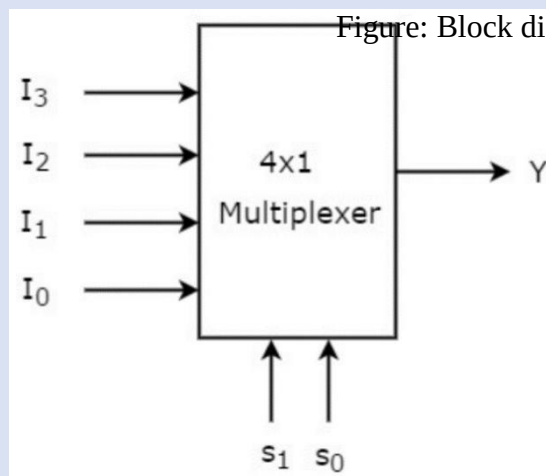


Figure: Block diagram and truth table of 2 to 1 MUX

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Figure: Block diagram and truth table of 4 to 1 MUX

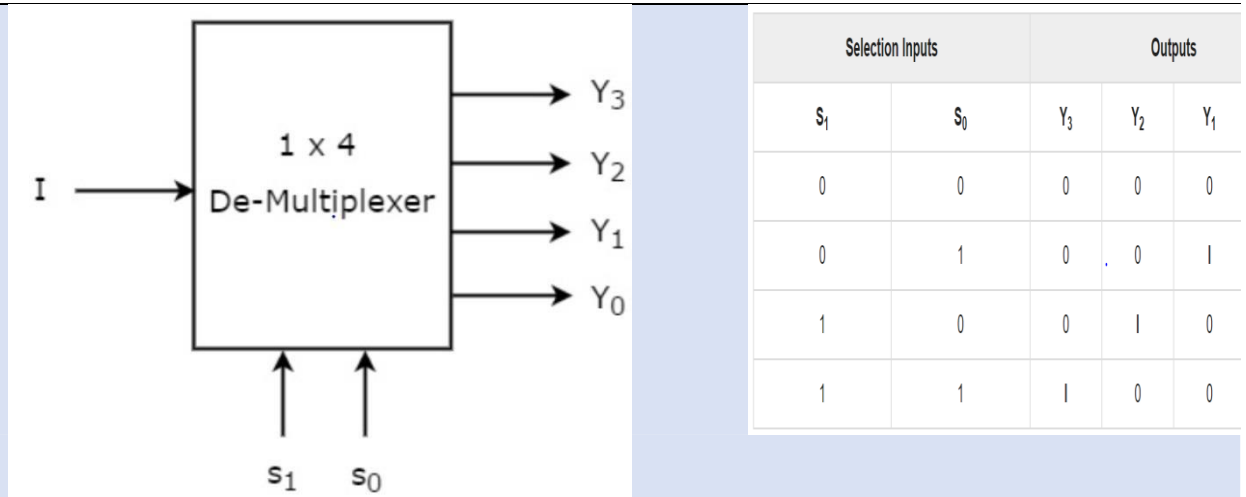


Figure: Block diagram and truth table of 1 to 4 Demultiplexer

Logic Diagram:

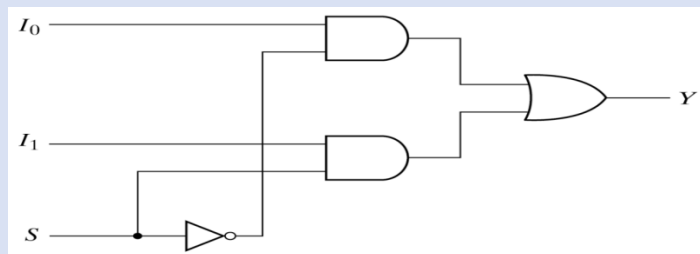


Figure: Logic diagram of 2 to 1 MUX

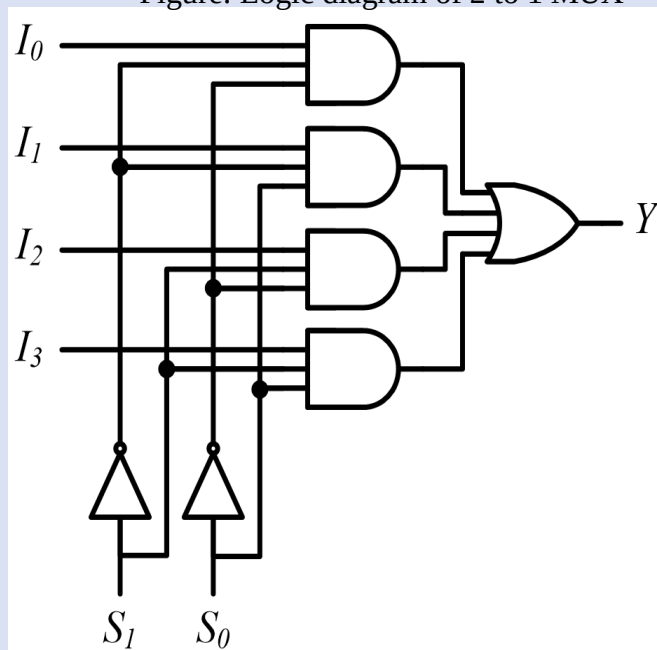


Figure: Logic diagram of 4 to 1 MUX

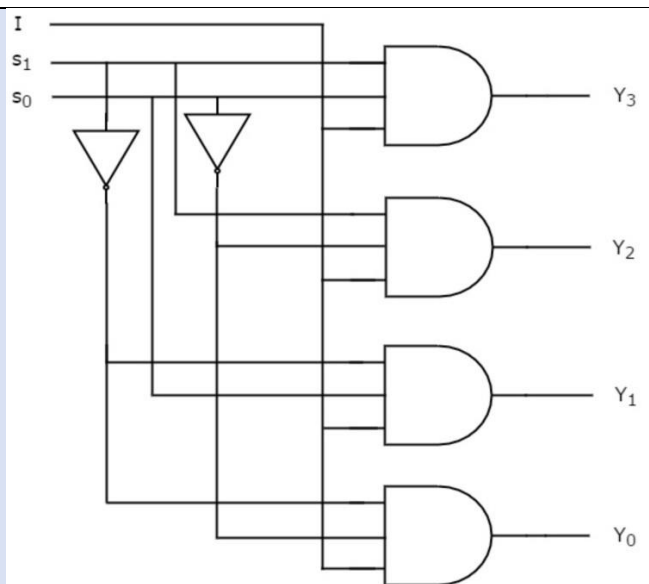


Figure: Logic diagram of 1 to 4 Demux

Procedure:

1. Check all the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.
4. Verify the Truth Table and observe the outputs.

Discussion:

1. Supply should not exceed 5v.
2. Connections should be tight and easy to inspect.
3. Use L.E.D. with proper sign convention and check it before connecting in circuit.

References:

1. Reference book.
2. Online sources.

30 Experiment
No.: 11

Experiment No. 11

Name of the Experiment: Designing of 4-BIT Magnitude comparator and magnitude comp using 74LS85 IC.

Objective: To learn about various applications of comparator. To learn and understand the w of IC 7485 magnitude comparator.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO3:Design and implement combinational circuit.	Design of Solutions (PO3)	Psychomotor/Guided response	☒ Simulation	☒ Lab tests ☒ Lab reports

COMPARING INPUTS				OUTPUT		
A3, B3	A2, B2	A1, B1	A0, B0	A > B	A < B	A = B
A3 > B3	X	X	X	H	L	L
A3 < B3	X	X	X	L	H	L
A3 = B3	A2 > B2	X	X	H	L	L
A3 = B3	A2 < B2	X	X	L	H	L
A3 = B3	A2 = B2	A1 > B1	X	H	L	L
A3 = B3	A2 = B2	A1 < B1	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 > B0	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 < B0	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	H

H = High Voltage Level, L = Low Voltage Level, X = Don't Care

Figure: Block diagram and truth table of 2 to 1 MUX

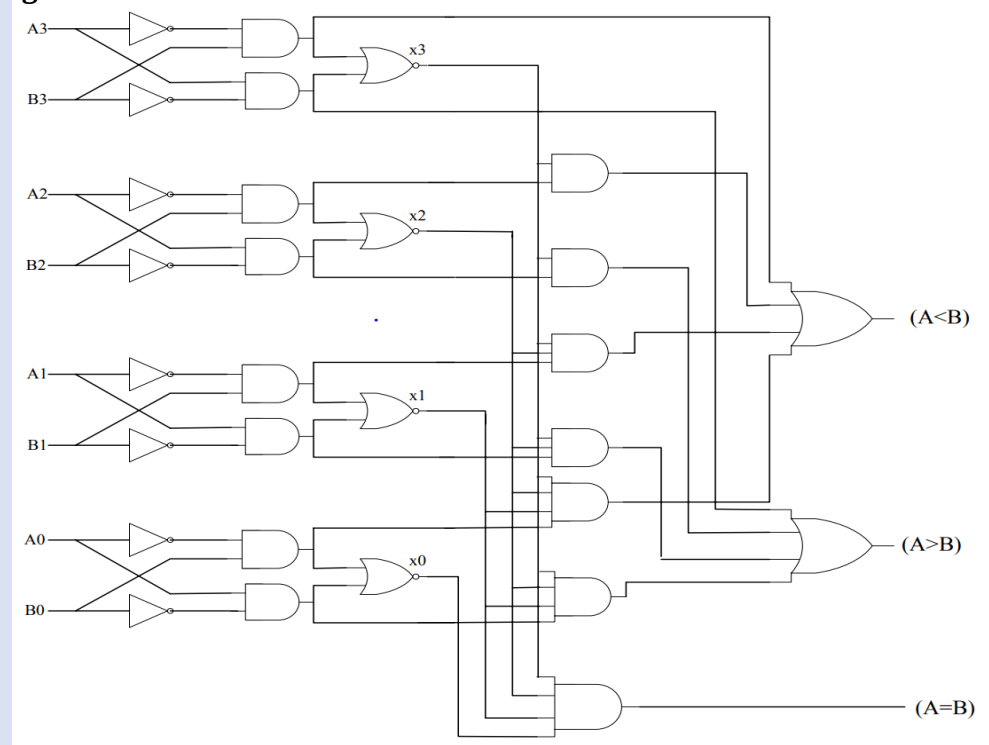
Logic Diagram:

Figure: Logic diagram of 4 bit magnitude comparator

Procedure:

5. Check all the components for their working.
6. Insert the appropriate IC into the IC base.
7. Make connections as shown in the circuit diagram.
8. Verify the Truth Table and observe the outputs.

Discussion:

4. Supply should not exceed 5v.
5. Connections should be tight and easy to inspect.
6. Use L.E.D. with proper sign convention and check it before connecting in circuit.
7. Four bit comparators are verified using basic gates and magnitude comparator IC7485.

References:

1. Reference book.
2. Online sources.

**31 Experiment
No.: 12**
Experiment No. 12

Name of the Experiment: Construct, test and investigate the operation of various types of Latches

Objective: To study and verify the truth table of logic gates of different types of Latches and identify various ICs and their specification

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO3: Design and implement combinational and sequential circuit.	Design of Solutions (PO3)	Psychomotor/Guided response	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Components Required:

Logic gates (IC) trainer kit. Digital Lab Kit, Breadboard, Power Supplier, Connecting patch cord
7400, IC 7408, IC 7432, IC 7406, IC 7402, IC 7404, IC 7486

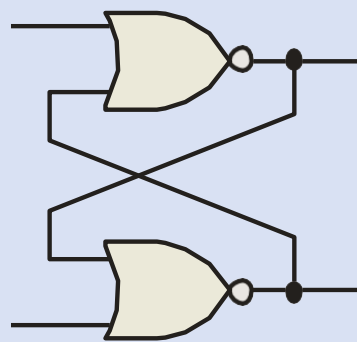
Theory:

A latch is a temporary storage device that has two stable states (bi-stable). It is a basic form of memory. The S-R (Set-Reset) latch is the most basic type. It cannot be constructed from NOR gates or NAND gates. With NOR gates, the latch responds to active-HIGH inputs; with NAND gates, it responds to active-LOW inputs.

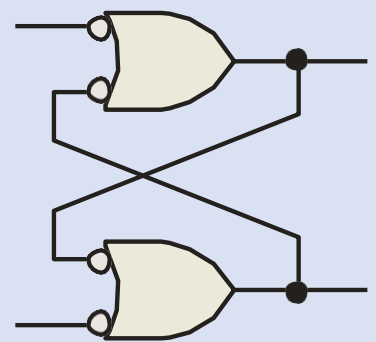
SR Latch:

An S-R latch consists of two cross-coupled NOR gates. An S-R flip-flop can also be designed using cross-coupled NAND gates as shown. The truth tables of the circuits are shown below. A clocked S-R flip-flop has an additional clock input so that the S and R inputs are active only when the clock is high. When the clock goes low, the state of flip-flop is latched and cannot change until the clock goes high again. Therefore, the clocked S-R flip-flop is also called "enabled" S-R flip-flop.

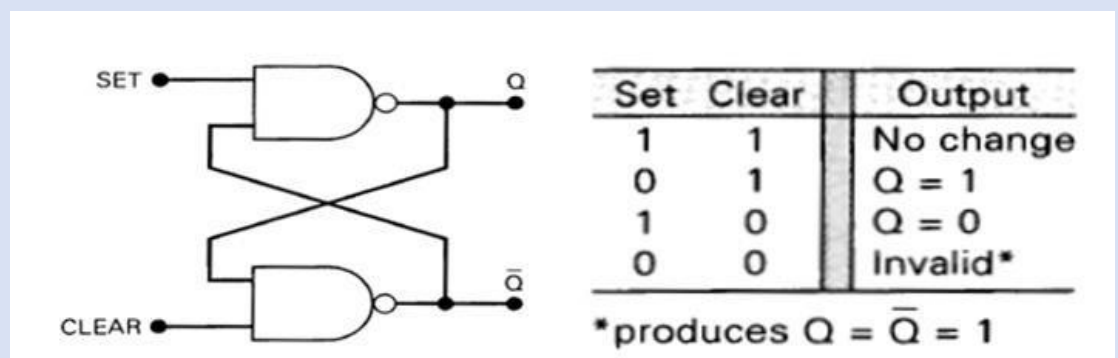
A D latch combines the S and R inputs of an S-R latch into one input by adding an inverter. When the clock is high, the output follows the D input, and when the clock goes low, the state is latched.



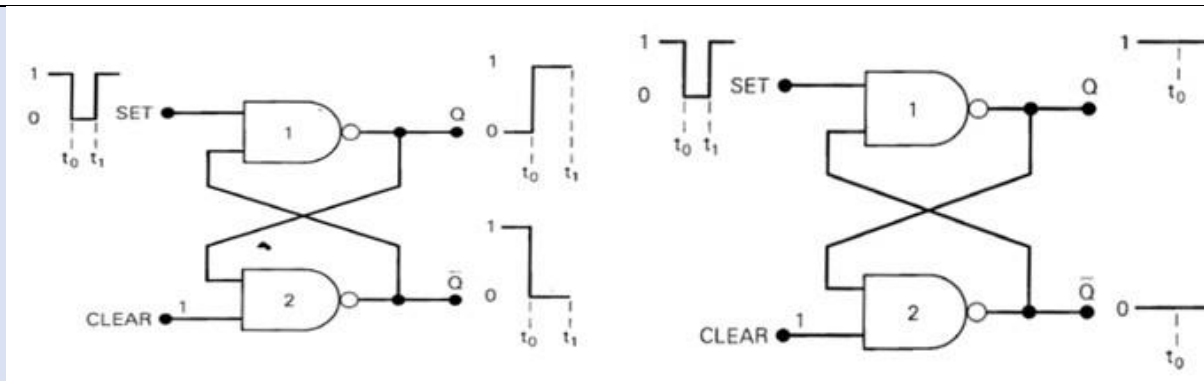
NOR Active-HIGH Latch



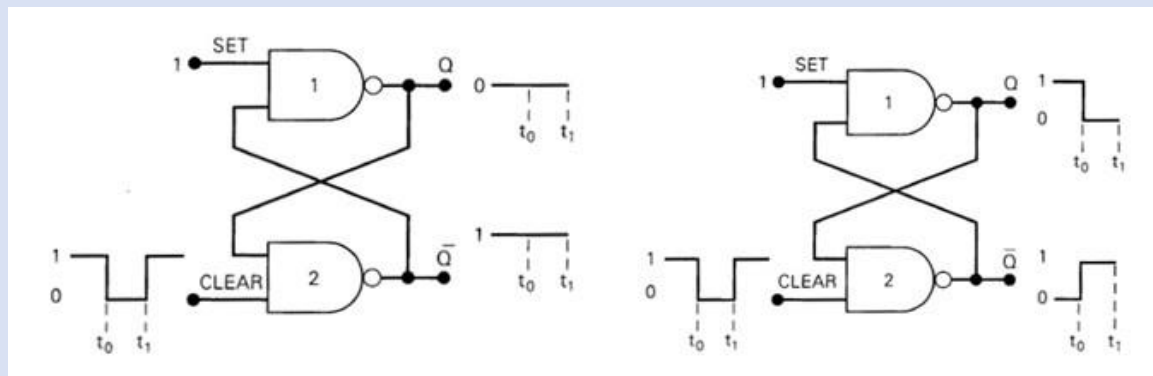
NAND Active-LOW Latch



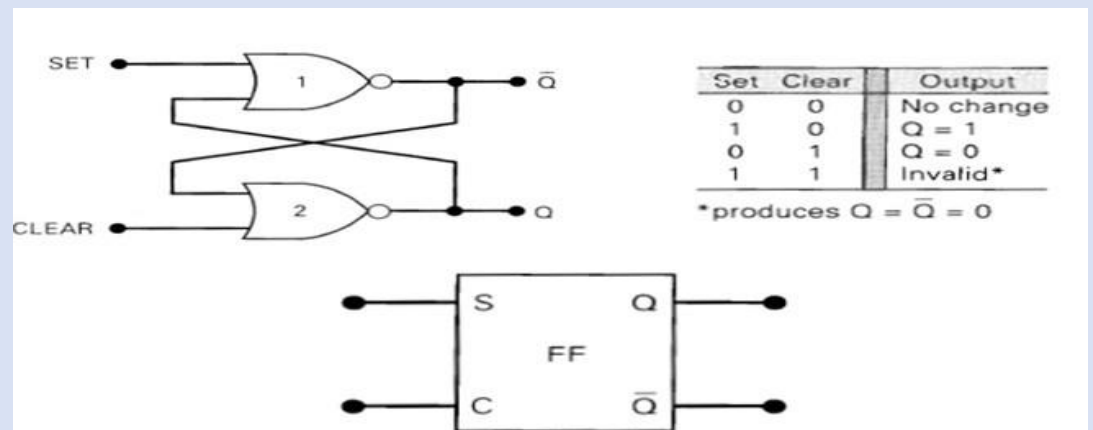
NAND Gate Latch with Truth Table



Setting NAND Gate Latches



Resetting NAND Gate Latches



NOR Gate Latch with Truth Table

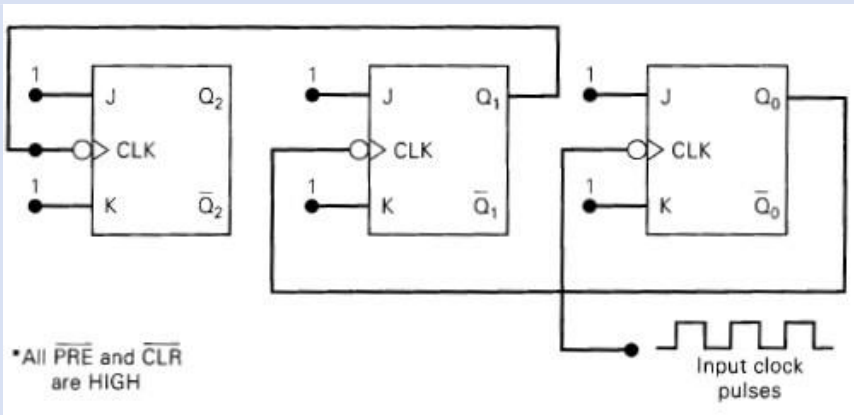
PROCEDURE:

- Check all the components for their working.
- Insert the appropriate IC into the IC base.
- Make connections as shown in the circuit diagram.
- Verify the Truth Table and observe the outputs.

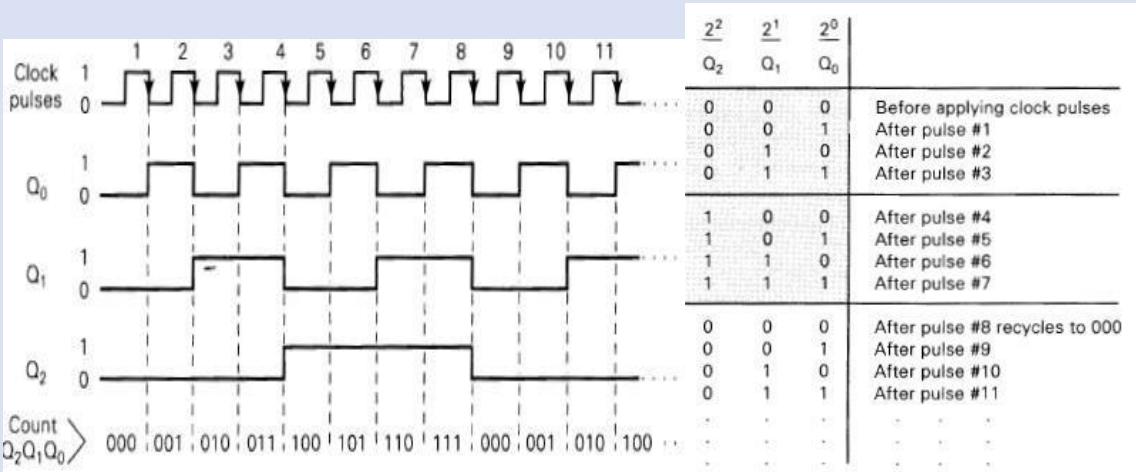
Discussion:

		Here we learned to Construct SR Latch using NOR and NAND gates, Setting and Resetting SR L and observed their Characteristics. References: 1. Reference book. 2. Online sources.										
32	Experiment No.: 13	<u>Experiment No. 13</u> <u>Name of the experiment:</u> Construct, test and investigate the operation of various types of Counters. <u>Objective:</u> To study and verify the truth table of logic gates of different types of Counter identify various ICs and their specification. <u>Course Outcomes (COs), Program Outcomes (POs) and Assessment:</u> <table><tr><th>CO Statement</th><th>Corresponding PO</th><th>Domain / level of learning taxonomy</th><th>Delivery methods and activities</th><th>Assessment tools</th></tr><tr><td>CO3: Design and implement combinational and sequential circuit.</td><td>Design of Solutions (PO3)</td><td>Psychomotor/Guided response</td><td> ☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial </td><td> ☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation </td></tr></table> Components Required: Logic gates (IC) trainer kit. Digital Lab Kit, Breadboard, Power Supplier, Connecting patch chord 7400, IC 7408, IC 7432, IC 7406, IC 7402, IC 7404, IC 7486 Theory: Ring counter is a basic register with direct feedback such that the contents of the register circulate around the register when the clock is running. Here the last output that is QD in a shift register is connected back to the serial input.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools	CO3: Design and implement combinational and sequential circuit.	Design of Solutions (PO3)	Psychomotor/Guided response	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation
CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools								
CO3: Design and implement combinational and sequential circuit.	Design of Solutions (PO3)	Psychomotor/Guided response	☒ Simulation ☒ Experiment ☒ Practice lab ☒ Group discussion ☒ Tutorial	☒ Lab tests ☒ Lab reports ☒ Final lab test ☐ Open ended lab ☐ Project show & project presentation								

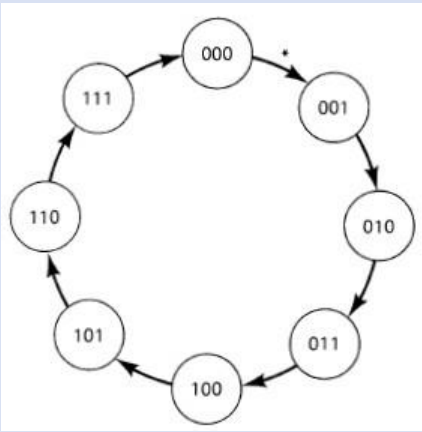
A basic ring counter can be slightly modified to produce another type of shift register counter Johnson counter. Here complement of last output is connected back to the not gate input and next output is connected back to serial input. A four bit Johnson counter gives 8 state output.



Frequency Division



Frequency Division



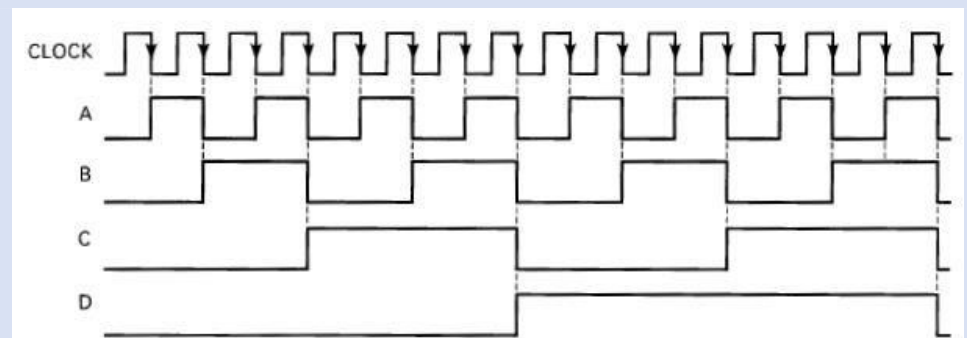
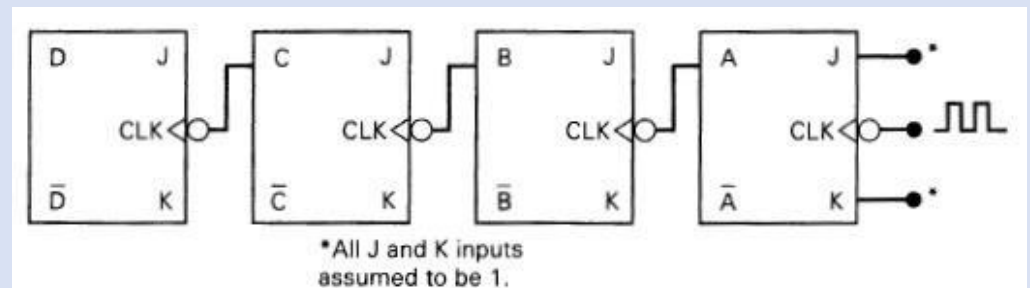
State Transition Diagram

MOD Number

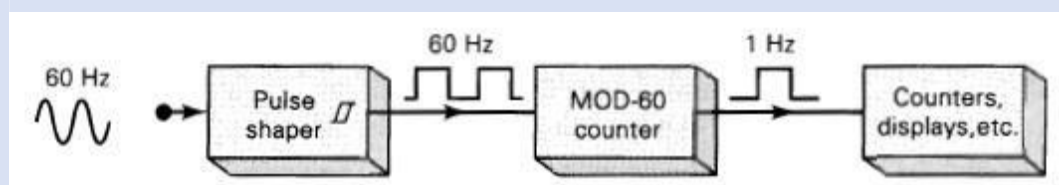
- MOD Number indicates number of states in counting sequence
- Last diagram: MOD number= $2^3 = 8$
- Consider a counter circuit that has six flip flops wired together as in last diagram.
 - a. What is the MOD number?
 - b. What is the frequency of last FF output if input clock has a frequency of 1 MHz?
 - c. What is the range of counting states of this counter?
 - d. Assume a starting state of (000000). What will be the counter's sequence after pulse?

Counters

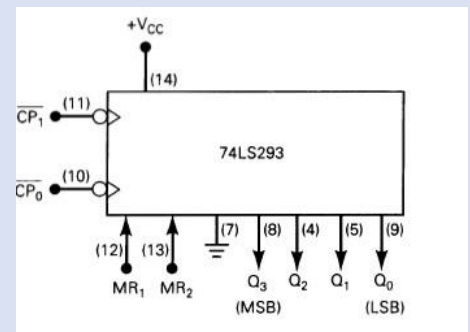
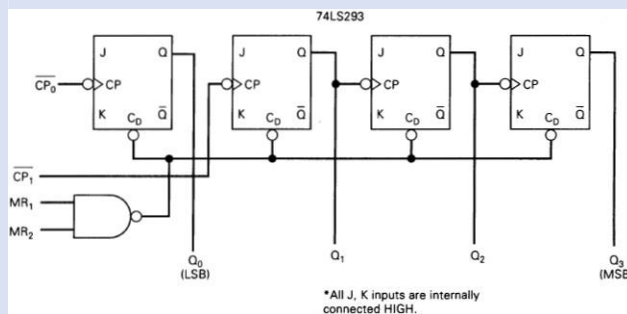
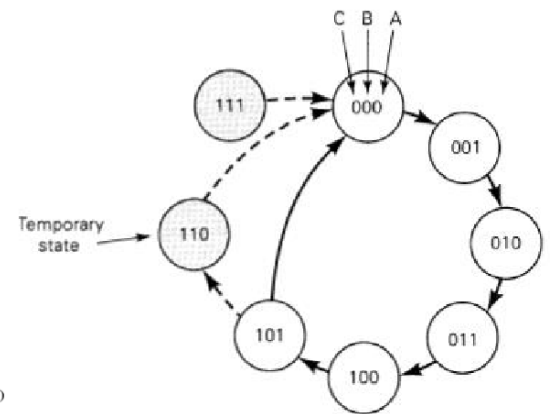
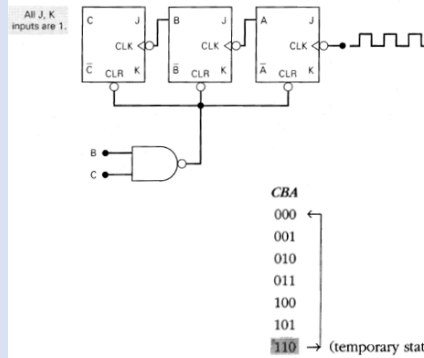
- Asynchronous (Ripple Counter)
 - Up Counter
 - Down Counter
- Synchronous (Parallel Counter)



Ripple Counter

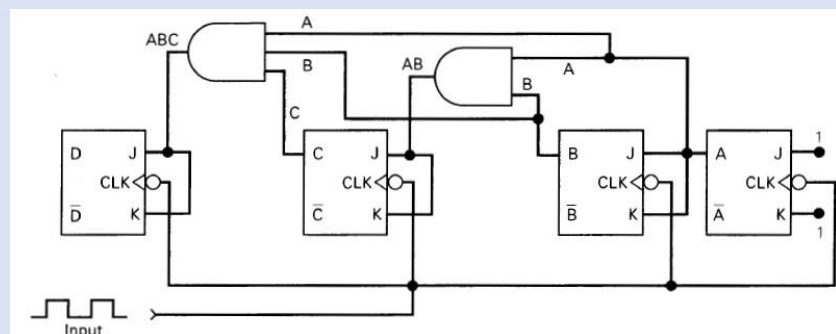
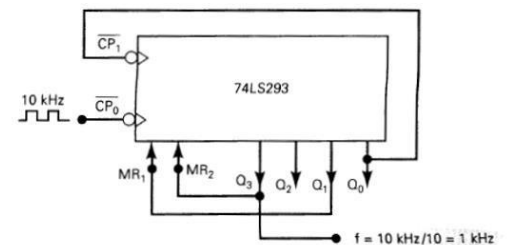
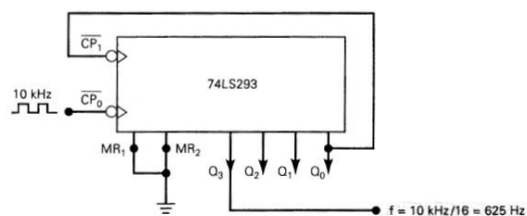


Example



IC Ripple Counter

- Design a MOD-16 counter using 74LS293 with a 10KHz clock input
- Design a MOD-10 counter using 74LS293 with a 10KHz clock input



Synchronous/Parallel Counter

Synchronous/Parallel Counter

Count	D	C	B	A
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
0	0	0	0	0
.	.	.	.	.
.	.	.	.	.
.	.	etc.	.	.

Basic Principle of designing a synchronous counter

Each FF should have its J and K inputs connected such that they are only HIGH when the outputs of all lower order FF are in HIGH state

Truth Table

Basic Principle of designing a synchronous counter:

Each FF should have its J and K inputs connected such that they are only HIGH when the output lower order FF are in HIGH state

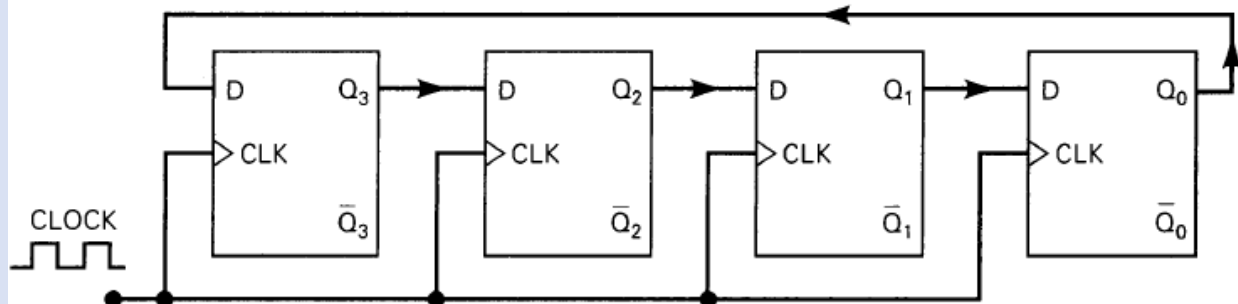
Shift Register Counter

- RING Counter
- Johnson Counter

4 Bit Ring Counter

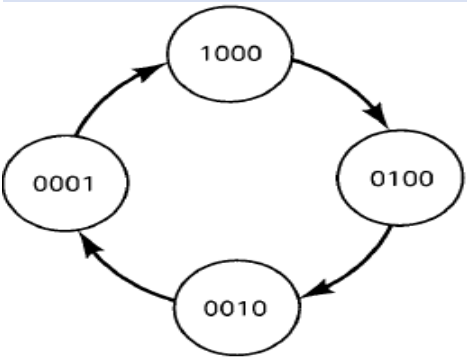
The ring counter can also be implemented with either D flip-flops or J-K flip-flops.

Here is a 4-bit ring counter constructed from a series of D flip-flops. Notice the feedback.

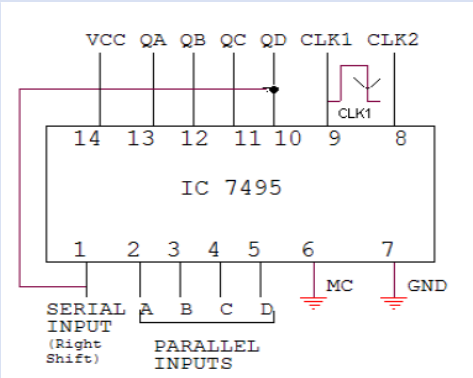


Q ₃	Q ₂	Q ₁	Q ₀	CLOCK pulse
1	0	0	0	0
0	1	0	0	1
0	0	1	0	2
0	0	0	1	3
1	0	0	0	4
0	1	0	0	5
0	0	1	0	6
0	0	0	1	7
.	.	.	.	.
.	.	.	.	.

Truth Table



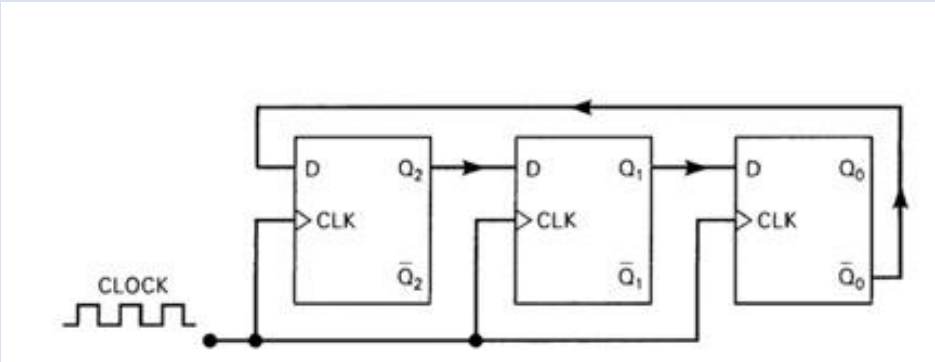
State Transition Diagram



Ring Counter Using IC 7495

Clock pulses	Q _A	Q _B	Q _C	Q _D
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	0	0
5	0	1	0	0
6	0	0	1	0
7	0	0	0	1
8	1	0	0	0

Truth Table

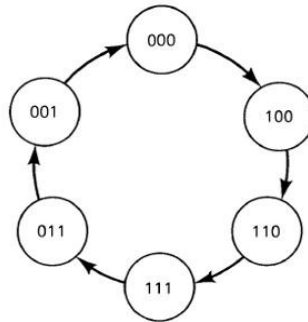


MOD-6 Johnson Counter

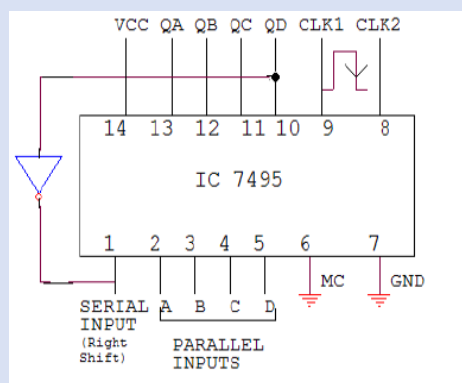
The Johnson counter is useful when you need a sequence that changes by only one bit at a time and has a limited number of states ($2n$, where n = number of stages).

MOD-6 Johnson Counter

Q ₂	Q ₁	Q ₀	CLOCK pulse
0	0	0	0
1	0	0	1
1	1	0	2
1	1	1	3
0	1	1	4
0	0	1	5
0	0	0	6
1	0	0	7
1	1	0	8
.	.	.	.
.	.	.	.
.	.	.	.



Q ₂	Q ₁	Q ₀	CLOCK pulse
0	0	0	0
1	0	0	1
1	1	0	2
1	1	1	3
0	1	1	4
0	0	1	5
0	0	0	6
1	0	0	7
1	1	0	8
.	.	.	.
.	.	.	.
.	.	.	.



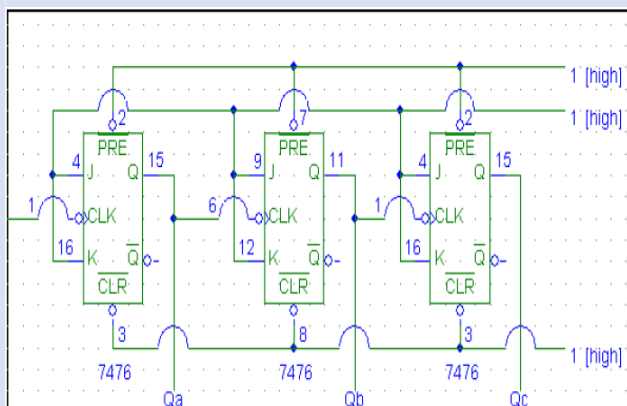
Clock pulses	Q _A	Q _B	Q _C	Q _D
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

Johnson Counter Using IC 7495

Truth Table

MOD-8 UP COUNTER

CIRCUIT DIAGRAM:



TRUTH TABLE

CLK	Q _C	Q _B	Q _A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Procedure:

- Check all the components for their working.
- Insert the appropriate IC into the IC base.
- Make connections as shown in the circuit diagram.

- Apply clock to pin number 9 and observe the output

Discussion: Here we learned to Construct Ripple UP/Down Counter, Ring Counter, Johnson C MOD-6 based Counter, MOD-8 Based Counter using frequency division techniques and MOD Techniques with different ICs and Basic gates and observed their Characteristics.

References:

1. Reference book.
2. Online sources.